POLYNOMIAL BEHAVIOUR OF INVARIANTS AFTER TENSORING WITH A POLYNOMIAL ALGEBRA OVER THE INVARIANTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For a polynomial algebra E over R^G the invariants of E tensor_{R^G} R under G are exactly E. invoked at proof lines 32648-32649. Identifies the base case where the lemma applies.
- The invariants modulo the ideal generated by J behave as described by the invariants-modulo-bis lemma, giving the polynomial identities for the quotient. invoked at proof lines 32652-32653. Transfers the polynomial identities from E to A = E/J.

Source reference. Original lemma 0H35: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32603–32603. Cited at proof lines 32652-32653 as the lemma of which the statement is a special case; the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let G be a finite group of order n acting on R. Let A be an R^G-algebra.*

(1) *for $b \in (A \otimes_{R^G} R)^G$ there exists a monic polynomial $P \in A[T]$ whose image in $(A \otimes_{R^G} R)^G[T]$ is $(T - b)^n$,*
(2) *for $a \in \mathrm{Ker}(A \to (A \otimes_{R^G} R)^G)$ we have $(T - a)^n = T^n$ in $A[T]$.*

Proof. Choose a surjection $E \to A$ where E is a polynomial algebra over R^G. Then $(E \otimes_{R^G} R)^G = E$ because E is free as an R^G-module. Denote $J = \mathrm{Ker}(E \to A)$. Since tensor product is right exact we see that $A \otimes_{R^G} R$ is the quotient of $E \otimes_{R^G} R$ by the ideal generated by J. In this way we see that our lemma is a special case of Lemma 0H35. □

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